

Office of the Provost
National University of Singapore
College of Design and Engineering
REx Fellows' Grant Application Form

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Note to Applicants:

- All fields must be completed. Incomplete forms may not be accepted.
 - Applicants are to submit the completed grant application form in **PDF format** online via the CDE Enhancement Portal by the deadline stipulated for each grant call.
 - For group projects, only one form is to be submitted per group. The form should be submitted on the CDE Enhancement Portal by the group leader.
 - All text should be single spacing, Arial, 10 pt.
 - For details on eligibility and terms and conditions, kindly refer to the REx Fellows' Grant Guidelines.
 - For general and administrative enquiries, please contact the UREx Administrative team at ugep@nus.edu.sg or contact your Programme's UREx coordinator. For claim related-queries, kindly contact Undergraduate Research Coordinating Team: John Caines (Manager) jlc@nus.edu.sg or Siti Nur Fairuz Sheikh Ismail (Instructor) fairuzsi@nus.edu.sg or Siti Fathima (Executive) sfathima@nus.edu.sg
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Research Grant Proposal

- (a) UREx project title: **Autonomous Electrical and Software Subsystems for the 2026 CanSat Competition**
- (b) Type of project: Individual / Group: Group
- (c) Abstract of UREx project (no more than 100 words)

The CanSat competition simulates a real satellite mission at miniature scale. This UREx project focuses on designing and developing the electrical and software subsystems for a CanSat that must autonomously perform telemetry, onboard sensing, flight-state control, and communications.

Our team's UREx role centers on designing the power system, integrating multiple onboard sensors (e.g., pressure, temperature, GPS, IMU), ensuring real-time telemetry transmission to a ground station via XBEE radios, and developing robust onboard flight software with fault-tolerant state transitions. These systems enable precise descent control, egg-release mechanism triggering, and full compliance with the 2026 CanSat mission requirements.

(d) Background literature and research hypothesis of UREx project

Mission Overview (2026)

The CanSat is launched inside a rocket and deployed at apogee (~725 m). After ejection, it descends with a parachute. At 80% of peak altitude, the payload separates and descends using a para-glider while steering toward a target zone. A simulated instrument (raw hen's egg) is released at 2 m altitude. Throughout the flight, the CanSat transmits sensor data at 1Hz to a ground station and records video from two onboard cameras.

The electrical and software subsystems are mission-critical in enabling this sequence to function autonomously.

Electrical Subsystem:

Key electrical tasks include:

- Designing the power architecture with runtime ≥ 2 hours using Li-ion (metal-cased) or NiMH batteries.
- Isolating power for the audio beacon, which must be independently powered.
- Sensor integration for temperature, pressure, GPS, IMU, magnetometer, and voltage/current monitoring.
- Configuring power switches, indicators, and connectors accessible through the container wall.
- Interfacing all sensors via I2C/UART/SPI protocols and ensuring hard mounting with vibration tolerance.

Software and Telemetry Subsystem:

Flight software governs autonomous operation via a **finite state machine (FSM)**:

Launch Pad → Ascent → Apogee → Payload Release → Descent → Instrument Drop → Landing

Key functionalities:

- Persistent packet counter and UTC mission time (even after resets)
- Real-time telemetry packet transmission (26+ fields) in CSV format at 1 Hz via XBEE
- Command handler for ground station commands (CX, CAL, ST, MEC, SIM)
- Simulation Mode support to inject altitude data using barometric CSV profiles
- Real-time mechanism activation via radio command (e.g., parachute deploy, egg release)
- Fail-safe recovery from power or processor resets

Ground software will be developed in Python to support:

- Real-time GUI for telemetry plots and sensor display
- Command terminal to issue control commands
- CSV data saving with proper header and packet tracking

(e) Brief description of methods and approaches

Design Approach:

- **Subsystem Architecture:** Define modular hardware/software blocks (sensors, radio, logic, power) to ensure ease in implementing isolated feedback loops.
- **Sensor Selection:** Evaluate IMUs, GPS modules, barometric sensors for accuracy, cost, integration ease.
- **Power System:** Separate main + beacon power rails; test power draw under load.
- **PCB Design (Optional):** Create integrated board using KiCad or breadboarded prototypes.
- **Software FSM:** Develop FSM logic using Arduino IDE or ESP32 C++ with watchdogs, timers, and interrupt routines.

Implementation:

- **Microcontroller:** Use Arduino Nano or ESP32 as a central flight computer.
- **Radio Comm:** Set up XBEE (900 MHz or 2.4 GHz) modules with proper NETID/PANID.
- **Camera Triggering:** Control camera through software-triggered relays or GPIO logic.
- **Ground Station Software:** Built using Python (Tkinter or PyQt) to display real-time telemetry, log data, and control flight.

Testing Plan:

- **Simulation Mode Testing:** Use mock CSV barometric data to validate FSM transitions and altitude triggers.
- **Drop + Thermal + Vibration Tests:** Validate sensor mounting and telemetry integrity under CanSat environmental test conditions.
- **Radio Range Testing:** Ensure >1 km telemetry reliability under field conditions.

Timeline

Month	Milestone
Aug-Sept 2025	Finalize subsystem roles, begin component testing
Oct–Nov 2025	Design architecture, draft PDR slides
Jan 2026	PDR submission and presentation
Feb 2026	Finalize design, order parts, begin CDR prep
Mar 2026	CDR submission and simulation mode testing
Apr–May 2026	Subsystem integration, testing, environmental tests
Jun 2026	Launch Weekend: flight readiness, launch, Post Flight Review

(f) Proposed budget breakdown and justification

Note: Applicant should discuss with UREx supervisor(s) on the overall proposed budget. Up to \$2,500 (for individual projects) or \$5,000 (for group projects) research grant will be provided to support the cost of research work and deliverables for UREx project. Applicants and potential supervisors may refer to REx Fellows' Grant Guidelines for details pertaining to the terms and conditions of the grant.

Item	Proposed budget (SGD)	Justification
Round Trip Plane Ticket	\$3000	The CanSat competition is held in the United States. A round trip flight from Singapore to Virginia for one student typically ranges from 2500 to 2800 SGD depending on the time of booking and airline availability. We are budgeting 3000 SGD to ensure coverage for airfare, transit, and essential travel expenses associated with attending the competition physically.

Competition Fee	\$270	The official registration fee for the competition is 200 USD per team, which converts approximately to 270 SGD. This fee is required for participation and covers launch equipment, rocket motors, and other logistical expenses managed by the organizers.
Sourcing of Materials, Sensors, Electronics	\$1700	The electrical and software systems require multiple components including XBEE radios, GPS modules, barometric and IMU sensors, microcontrollers, audio beacon systems, voltage regulators, camera modules, SD cards, and power management hardware. Additional costs will be incurred for soldering tools, breadboards, connectors, wires, batteries, voltage rails, and prototyping boards. To ensure redundancy, we will prototype multiple versions of the flight software and telemetry system and carry backup modules during launch. Integration and testing require extensive prototyping, and manufacturing spare subsystems is essential to ensure robustness under competition conditions. We estimate 1700 SGD for complete sourcing and testing of electronic and computing hardware required for successful flight.
	Total: \$4970	

(g) References and appendices (if any)

Urbanozo, A., Reed, A., Ruse, N., Jones, Z., Littlepage, J., Treadway, A., Chouinard, S., Teer, M., Slusser, H., Mickel, N., University of Alabama in Huntsville, & Space Hardware Club. (n.d.).
CanSAT 2024 PDR: Team #2078 Shockwave.

https://cansatcompetition.com/docs/teams/cansat2024_2078_pdr_v02.pdf

Cansat Competition Guide. (2026).

https://cansatcompetition.com/docs/CanSat_Mission_Guide_2026.pdf

CanSAT 2024 PDR: Team 2057 (AntSAT). (2024).

https://cansatcompetition.com/docs/teams/cansat2024_2057_pdr_v01.pdf



Signature of applicant

15th August 2025

Date

REx Fellows' Grant Guidelines (CDE)

1. Introduction

- 1.1 Students who are interested to enrol in UREx should submit an application form to be eligible for a research grant of up to \$2,500 (for individual projects) or \$5,000 (for group projects) which can be used to reimburse costs of the associated UREx project.
- 1.2 The research grant will be provided on a requirement basis for the UREx project. Each grant application form submitted should account for the expenses involved in the utilisation of the grant.

2. Eligibility Criteria

Students enrolling for UREx should satisfy the following criteria:

- Must be a registered student of CDE
- Have at least Stage 2 standing
- Have at least two regular semesters of study remaining prior to graduation
- UREx cannot be undertaken concurrently with UROP (EG2605), SEP, NOC, or Industrial Attachment (IA). Students must ensure they have submitted their final report for UREx before they commence UROP, SEP, NOC, or IA
- For students taking both UROP and UREx (Note: cannot be taken concurrently), the scope of the project for UROP should be different from that of UREx.
- FYP can be taken concurrently with UREx, however the FYP project should be significantly different from the UREx project
- UREx can only be taken once; students have a maximum of two regular semesters to complete UREx
- Main supervisor for UREx must be a full-time academic staff from CDE (includes staff with joint appointments with CDE).

3. Application Procedure and Approval of Project Grant

- 3.1 The call for REx grant proposals will be made twice a year. The grant call and closing dates will be communicated to applicants by the UG Research Coordinating team at the Office of the Provost (PVO).
- 3.2 Each grant proposal will be evaluated by the UG Research Coordinating team according to the following criteria:
 - (a) Clear articulation of motivations for seeking to enrol in UREx.
 - (b) Clear articulation of research gaps and proposed methods to address these gaps in the UREx.
 - (c) Justifiable proposed budget breakdown of REx research grant.
- 3.3 Applicants are to fill out all sections of the REx Fellows' Grant Application Form containing applicant's details, description of the project and a tabulated proposed budget breakdown. The completed form should be uploaded online via the Enhancement Portal as a PDF file named as "StudentName_Department_CDE_MonthYear.pdf", where "Department" refers to the department hosting the UREx project. For group projects, only one form is to be submitted per project.
- 3.4 Applicants will be informed of their application outcome by email. Upon approval of the grant application form, applicants will be named as a "REx Fellow" and will be allocated the UREx course (CDE2605R) on their CANVAS page.

4. Terms and Conditions

- 4.1 Supervisors have the responsibility to ensure that the project carried out by REx Fellows are in compliance with NUS's code of ethics, research integrity and responsible research conduct.
- 4.2 Wherever applicable, supervisors should ascertain that necessary approvals are in place before carrying out the research activities.
- 4.3 The REx Fellows' research grant shall be used solely for the purposes of the associated UREx project. REx Fellows have up until their final semester before graduation to utilise the grant monies.
- 4.4 The following items are allowed to be purchased using the REx Fellows' research grant:
- (a) Consumables, lab supplies, use of research facilities and equipment directly required for the associated UREx project.
 - (b) License fees of software to be used for the associated UREx project.
 - (c) Registration fees for conferences, seminars or workshops.
 - (d) Costs of preparing project deliverables such as costs of printing posters, travel and transportation costs directly associated to carrying out the associated UREx project.
 - (e) Books or journal articles which are directly related to the associated UREx project. Subscription to journals which are NOT available in the university's libraries is allowed.
 - (f) Fees associated with publications of manuscript in journals.
- 4.5 The ownership of items purchased using the REx Fellows' research grant will be subject to agreements between REx Fellows and their supervisors. Items such as books, journal articles or posters can be kept by REx Fellows with the approval of their supervisors. Items to be used in a lab setting should be kept by the research lab.
- 4.6 The following items are not allowed to be claimed from the REx Fellows' research grant:
- (a) Entertainment and refreshment.
 - (b) Personal expenses not related to associated UREx project.
 - (c) Equipment or devices not related to associated UREx project.

5. Deliverables

The deliverables for UREx include:

- a) Attendance for a minimum of five REx workshops, two of which are core REx workshops - Ethics in Research, and Research Methodologies.
- b) — UREx report: 10-20 page technical report on the conducted research.
- e) — Executive summary: 1-2 page non-technical report that summarizes the conducted research in clear and concise layperson terms.

For group projects, each team member is required to complete the attendance of workshops and submit an individual UREx report and executive summary.